

Transcoronary Concentration Gradient of microRNA-133a and Outcome in Patients With Coronary Artery Disease



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Circulating levels of microRNA (miR)-133a are increased in patients with coronary atherosclerotic disease (CAD). Whether the cardiac release of this miR provides any prognostic information in patients with CAD is currently unknown. We aimed to investigate if changes in concentration of miR-133a through the coronary circulation may be associated with patients' cardiovascular outcome. We enrolled 111 patients (82 with stable CAD and 29 with acute coronary syndromes [ACS]) who underwent coronary angiography. Circulating levels of miR-133a were measured across the transcoronary circulation. Major adverse cardiac events (MACE: death, nonfatal myocardial infarction, and need for revascularization) were recorded through a median follow-up of 32 months. An increased transcoronary concentration gradient of miR133a showed a significant association with overall rate of MACE at follow-up in patients with both stable CAD and ACS ($p = 0.011$ and $p = 0.002$, respectively). At the single end point-analysis, increased transcoronary concentration gradients of miR133a were significantly associated with increased rate of death in patients with ACS ($p = 0.017$) and with increased incidence of new revascularization because of in-stent restenosis in patients with stable CAD ($p = 0.026$). Kaplan-Meier curves showed a significantly worse event-free survival in patients with greater transcoronary gradients of miR133a ($p = 0.026$ in stable CAD group and $p = 0.007$ for ACS group). Nevertheless, these findings lost their significance after adjustment for common cardiovascular risk factor and high-sensitivity troponin-T. In conclusions, the release of miR133a, as measured by its transcoronary concentration gradient, is associated with a higher incidence of MACE in patients with CAD, but it does not add significant prognostic information compared with traditional prognostic biomarkers, therefore limiting its potential usefulness in the clinical practice. Nevertheless, the differential modulation of miR-133a release in the coronary circulation may reflect pathophysiological mechanism involved in CAD progression and complications and suggest a novel potential role for this miR in the development of in-stent restenosis. © 2017 Elsevier Inc. All rights reserved. (Am J Cardiol 2017;120:15–24)

MicroRNAs (miRs) are present in the human circulation in a remarkably stable cell-free form and their plasmatic levels are differently regulated in many pathological conditions, opening attractive possibilities to use circulating miRs as disease biomarkers.¹ The miR133 family is composed of

muscle-enriched miRs (miR133a/b), initially identified in cardiac and skeletal muscle, where they play an essential role in cellular proliferation, differentiation, survival, and hypertrophy.² More recently, it has been demonstrated that miR133 family members are also expressed in vascular smooth muscle cells (VSMCs), where they regulate VSMCs phenotypic switch in response to vascular injury, suggesting a pivotal role for this miR in development of pathological vascular remodeling and opening intriguing scenarios for the use of miR133 in diagnosis and treatment of vasculoproliferative diseases.³ Plasmatic concentrations of miR-133a are increased in patients with stable coronary atherosclerotic disease (CAD) and acute coronary syndromes (ACS). However, the potential prognostic value of miR-133a in these patients still remain largely unknown, as previous studies reported conflicting results.^{4–7} A potential confounding factor contributing to this inconclusive evidence is that miR-133a is not exclusively expressed in cardiac myocytes. As atherosclerosis is a systemic disease, alterations in plasma concentrations of miR-133a may not only be related to alterations in the coronary circulation but rather reflect disease processes in the systemic circulation.⁸ Therefore, we investigated whether changes in plasma concentrations of miR-133a through the coronary

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Table 1
Baseline patients' characteristics

Variable	Overall Population (N=111)	Stable CAD (N=82)	ACS (N=29)	p
Age (years)	63,4±11,3	63,5±11,7	63,4±10,3	0,96
Sex (M/F)	89/22	64/18	25/4	0,42
Arterial hypertension	93 (84%)	68 (56%)	25 (7%)	0,78
Diabetes Mellitus	29 (26%)	21 (17%)	8 (2%)	0,81
Current smoker	43 (39%)	32 (26%)	11 (3%)	0,88
Previous AMI	44 (40%)	37 (30%)	7 (2%)	0,051
Previous PCI	62 (56%)	55 (45%)	7 (2%)	<0,001
LVEF (%)	52,7±10,9	55,0±9,5	46,6±12,3	0,001
NYHA Class (0/I/II/III/IV)	21/39/32/16/3	16/31/19/14/2	5/8/13/2/1	0,21
CCS Class (0/I/II/III/IV)	22/8/14/28/39	22/8/14/27/11	0/0/0/1/28	<0,001
Leukocytes (x10 ⁹ /L)	8,2±2,9	7,5±2,1	10,5±3,9	<0,001
Serum creatinine (mg/dL)	1,0±0,2	0,99±0,22	1,1±0,3	0,41
hs-TnT _[CVS] (ng/mL)	167,6±593,0	9,9±7,7	580,9±1030,3	<0,001
NT-pro-BNP _[CVS] (pg/mL)	511,8±725,2	330,5±384,3	999,6±1116,6	0,003
C-reactive protein (mg/dL)	0,72±1,8	0,42±0,82	1,56±3,22	<0,001
Cholesterol (mg/dL)	185,3±46,6	185,4±46,7	185,1±47,1	0,98
-LDL-Cholesterol (mg/dL)	106,6±40,8	104,9±41,8	112,6±37,2	0,26
Statin	76 (68%)	66 (54%)	10 (3%)	<0,001
ACE-I or ARB	86 (77%)	71 (58,2)	15 (4%)	0,002
Insulin	10 (9%)	6 (5%)	4 (1%)	0,28
Antiplatelet therapy				
- Aspirin	94 (85%)	70 (57%)	24 (7%)	0,47
- Clopidogrel	55 (49%)	41 (34%)	14 (4%)	0,90
- Ticagrelor/Prasugrel	22 (20%)	14 (11%)	8 (2%)	0,28
Oral anticoagulation therapy	5 (4%)	5 (4%)	0 (0%)	0,35

All p values represent comparisons between CAD and ACS. Comparisons between groups were performed with the Fisher's exact test or chi-square test for categorical variables and with Student *t* test or Mann-Whitney *U* test for continuous variables. Bold indicates statistical significance at *p* < 0.05.

ACE-I = angiotensin-converting enzyme inhibitor; ACS = acute coronary syndrome; AMI = acute myocardial infarction; ARB = angiotensin receptor blocker; CAD = coronary artery disease; CCS = Canadian Cardiovascular Society; CVS = coronary venous sinus; hs-TnT = high sensitivity troponin T; LDL = low-density lipoprotein; LVEF = left ventricular ejection fraction; NT-pro-BNP = N-terminal pro-B-type natriuretic peptide; NYHA = New York Heart Association; PCI = percutaneous coronary intervention.

circulation, as measured by its transcoronary concentration gradient, relates to outcome in patients with CAD.

Methods

Patients undergoing coronary angiography in the catheterization laboratory of the Goethe University Hospital of Frankfurt were enrolled. Patients were classified into 2 groups as follows: patients with angiographically documented CAD (at least 1 lesion >50% of vessel lumen or previous coronary revascularization) and absence of high-sensitivity troponin T (hs-TnT) elevation >50pg/ml ("stable CAD group"); patients with angiographically documented CAD, clinical presentation and/or, electrocardiographic changes suggesting acute myocardial ischemia and elevation of hs-TnT >50pg/ml ("ACS group").⁹ Exclusion criteria included conditions that could affect the concentration of circulating miRs independently from the cardiovascular status (severe hepatic or renal dysfunction, ongoing malignant or inflammatory disease, history of leukopenia, or thrombocytopenia). Patients with myocarditis or cardiogenic shock were also excluded. The ethics review board of the Goethe University Frankfurt approved the protocol, and the study was conducted in accordance with the Declaration of Helsinki. Written informed consent was obtained from each individual.

Blood samples were simultaneously obtained from the coronary venous sinus (CVS) and from the aortic bulb

Table 2
Events at follow-up

Variable	Total (N=111)	Stable CAD (N=82)	ACS (N=29)	p
MACE	27 (24%)	16 (19%)	11 (38%)	0,07
Death	10 (9%)	4 (5%)	6 (21%)	0,019
Non-fatal-MI	3 (3%)	3 (4%)	0 (0%)	0,56
Revascularization	20 (18%)	14 (17%)	6 (21%)	0,78
-TLR (Restenosis)	9 (8%)	6 (7%)	3 (10%)	0,69
-de novo Stenosis	14 (13%)	10 (12%)	4 (14%)	0,98

ACS = acute coronary syndrome; CAD = coronary artery disease; MACE = major adverse clinical events; MI = myocardial infarction; TLR = target lesion revascularization.

All p values represent comparisons between CAD and ACS. Comparisons between groups were performed with the Fisher's exact test. Bold indicates statistical significance at *p* < 0.05.

through 5F catheters during the cardiac catheterization procedure, as previously described.¹⁰ Taking into account the well-known influence of heparin on miR measurements,¹¹ blood samples were obtained before heparin was administered. Samples were processed as previously described,¹⁰ and miR133a expression was detected by quantitative polymerase chain reaction (PCR) using the specific TaqMan microRNA Assay kits (ABI, Foster City, California).

Coronary angiography and percutaneous coronary intervention were performed in accordance to the current

Table 3
Events at follow-up according to miR-133a concentration gradients

Variable	miR-133a _[CVS]				miR-133a _[Ao]			miR-133 _[TCG]		
	Total (N=111)	< median (N=55)	> median (N=56)	p	< median (N=55)	> median (N=56)	p	< median (N=55)	> median (N=56)	P
A) Overall population										
MACE	27 (24%)	10 (18%)	17 (30%)	0,184	11 (20%)	16 (29%)	0,377	4 (7%)	23 (41%)	<0,001
Death	10 (9%)	2 (4%)	8 (14%)	0,094	3 (5%)	7 (12%)	0,321	0 (0%)	10 (18%)	0,001
Non-fatal-MI	3 (11%)	1 (2%)	2 (4%)	0,507	1 (2%)	2 (4%)	0,507	1 (2%)	2 (4%)	0,507
Revascularization	20 (18%)	8 (14%)	12 (21%)	0,460	9 (16%)	11 (20%)	0,806	4 (7%)	16 (29%)	0,006
-TLR (Restenosis)	9 (8%)	3 (5%)	6 (11%)	0,489	4 (7%)	5 (9%)	1,0	1 (2%)	8 (14%)	0,032
-de novo Stenosis	14 (13%)	6 (11%)	8 (14%)	0,776	5 (9%)	9 (16%)	0,394	4 (7%)	10 (18%)	0,151
B) Stable CAD patients										
Variable	Total (N=82)	< median (N=41)	> median (N=41)	p	< median (N=41)	> median (N=41)	p	< median (N=41)	> median (N=41)	P
MACE	16 (19%)	7 (17%)	9 (22%)	0,781	8 (19%)	8 (19%)	1,0	3 (7%)	13 (32%)	0,011
Death	4 (4%)	1 (2%)	3 (7%)	0,616	1 (2%)	3 (7%)	0,616	0 (0%)	4 (10%)	0,116
Non-fatal-MI	3 (4%)	0 (0%)	3 (7%)	0,241	1 (2%)	2 (5%)	0,500	1 (2%)	2 (5%)	0,500
Revascularization	14 (17%)	6 (15%)	8 (19%)	0,770	7 (17%)	7 (17%)	1,0	3 (7%)	11 (27%)	0,037
-TLR (Restenosis)	6 (7%)	3 (7%)	3 (7%)	1,0	3 (7%)	3 (7%)	1,0	0 (0%)	6 (15%)	0,026
-de novo Stenosis	10 (12%)	4 (10%)	6 (15%)	0,737	5 (12%)	5 (12%)	1,0	3 (7%)	7 (17%)	0,311
C) ACS patients										
Variable	Total (N=29)	< median (N=14)	> median (N=15)	p	< median (N=14)	> median (N=15)	p	< median (N=14)	> median (N=15)	P
MACE	11 (38%)	4 (29%)	7 (47%)	0,450	4 (29%)	7 (47%)	0,450	1 (7%)	10 (67%)	0,002
Death	6 (21%)	2 (14%)	4 (27%)	0,651	2 (14%)	4 (27%)	0,651	0 (0%)	6 (40%)	0,017
Non-fatal-MI	0 (0%)	0 (0%)	0 (0%)	-	0 (0%)	0 (0%)	-	0 (0%)	0 (0%)	-
Revascularization	6 (21%)	2 (14%)	4 (27%)	0,651	2 (14%)	4 (27%)	0,651	1 (7%)	5 (33%)	0,169
-TLR (Restenosis)	3 (10%)	2 (14%)	1 (7%)	0,598	2 (14%)	1 (7%)	0,598	1 (7%)	2 (13%)	0,500
-de novo Stenosis	4 (14%)	1 (7%)	3 (20%)	0,598	0 (0%)	4 (27%)	0,100	1 (7%)	3 (20%)	0,598

Ao = aorta; ACS = acute coronary syndrome; CAD = coronary artery disease; CVS = coronary venous sinus; MACE = major adverse clinical events; MI = myocardial infarction; TCG = transcatheter concentration gradient; TLR = target lesion revascularization.

All p values represent comparisons between patients with miR-133a concentrations below or above the median value. Comparisons between groups were performed with the Fisher's exact test. Bold indicates statistical significance at $p < 0.05$.

guidelines.¹² For selective cannulation of CVS, a left Amplatz 1 or 2 through the venous femoral access was used. Subsequent medical therapy was established in accordance to the current guidelines.^{13–15}

The primary end point was the occurrence of major adverse cardiovascular events (MACE), defined as a composite of death, nonfatal MI, and need for revascularization, at follow-up. Follow-up was performed either by outpatient visit or by telephonic interview.

All statistical calculations were performed using SPSS 22 (SPSS, Inc., Chicago, Illinois). Differences between groups were assessed using the Fisher's exact test or chi-square test for the categorical variables. For continuous variables with normal distribution, the Student *t* test was used, whereas for continuous variables without normal distribution, the nonparametric Mann-Whitney *U* test was performed. The Kaplan-Meier curves were used to illustrate the timing of events at follow-up; statistical assessment was performed using the log-rank test. Simple Cox regression analysis was performed to identify predictors of MACE. Significant variables were tested in a stepwise multiple Cox regression analysis. A *p* value < 0.05 was considered statistically significant.

Results

A total of 111 patients (82 patients with stable CAD and 29 patients with ACS) were enrolled. Baseline characteristics of enrolled patients are listed in Table 1. Both subgroup populations belonged to high-risk category according to common clinical scores (for stable CAD patients, ACEF [Age, Creatinine and Ejection Fraction] risk score¹⁶ mean: 1.3 ± 0.39 ; for ACS patients, GRACE [Global Registry of Acute Coronary Events] risk score¹⁷ mean: 130 ± 37.9). The median duration of follow-up was 32 months (interquartile range: 13 to 43 months). Dual antiplatelet therapy was prescribed according to the current guidelines. At follow-up, all patients were on optimal medical therapy. No patient had statin or acetylsalicylic acid intolerance. As listed in Table 2, a first MACE occurred in 27 (24.3%) patients. Ten patients (9%) died during follow-up. The causes of death were as follows: cardiac deaths (cardiogenic shock or re-MI, $n = 6$), cancer ($n = 1$), multiorgan failure from noncardiac causes ($n = 2$), and unknown ($n = 1$). After dividing patients into 2 groups according to the median miR-133a concentration in CVS, aorta, and in coronary

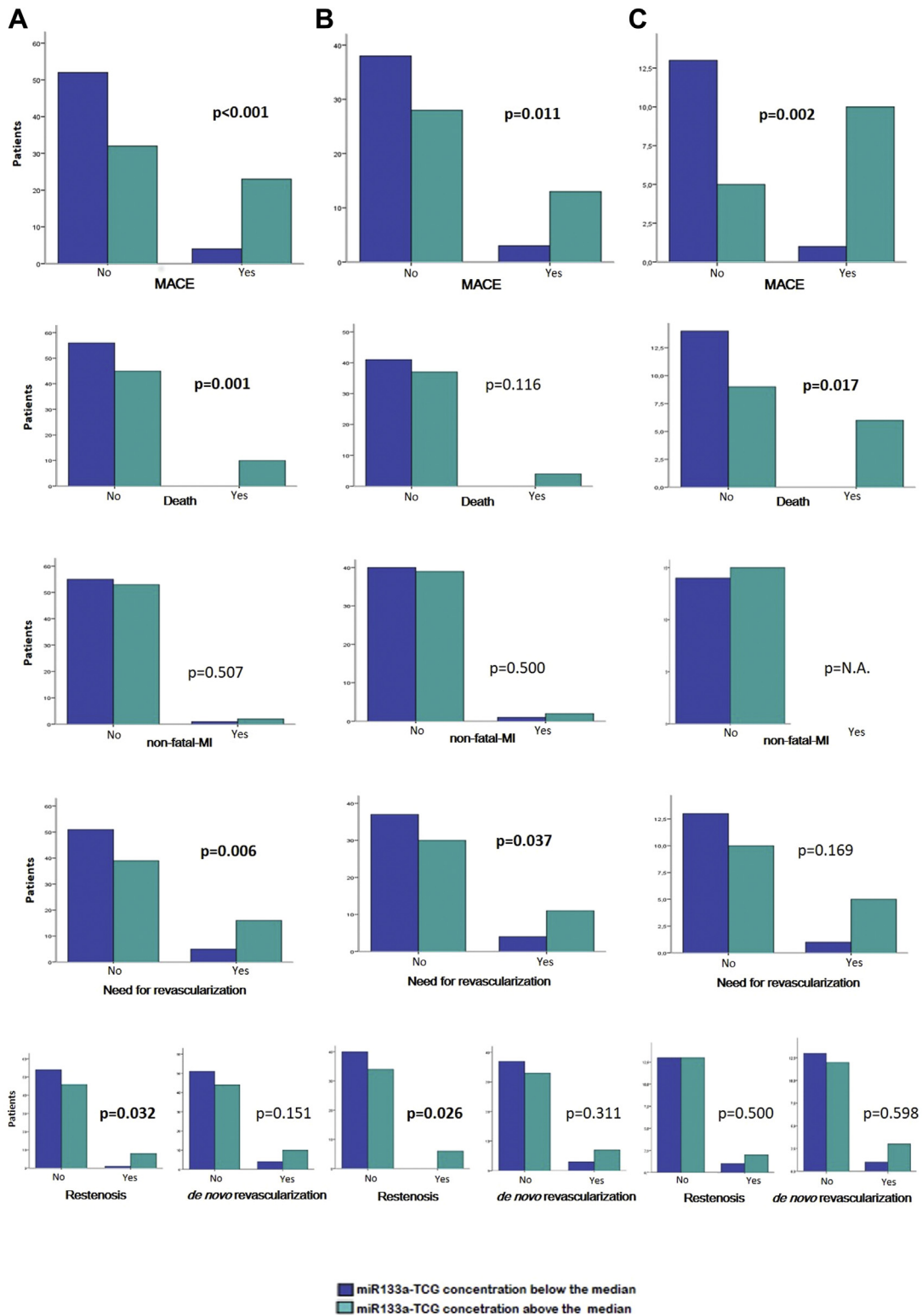


Figure 1. Transcoronary concentration gradients of miR-133a and MACE. (A) In the overall population, miR-133a transcoronary concentration gradients above the median were associated with a significantly higher incidence of cumulative MACE, death, and need for revascularization (driven by a significantly higher rate of target lesion revascularization because of in-stent restenosis) at follow-up; (B) in patients with stable CAD, miR-133a transcoronary concentration gradients above the median were associated with a significantly higher incidence of cumulative MACE and need for revascularization (driven by a significantly higher rate of target lesion revascularization because of in-stent restenosis) at follow-up; (C) in patients with ACS, miR-133a transcoronary concentration gradients above the median were associated with a significantly higher incidence of cumulative MACE and death at follow-up. Differences between groups were assessed using the Fisher's exact test.

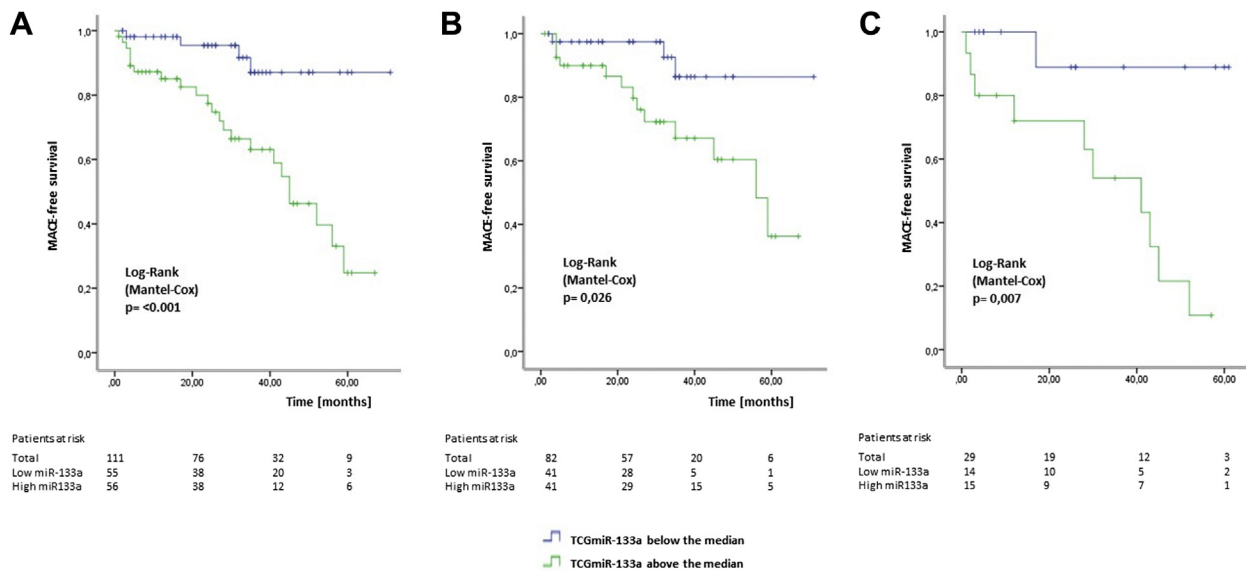


Figure 2. MACE-free survival according to transcoronary concentration gradients of miR-133a. Patients with transcoronary concentration gradients of miR-133a below the median showed a significantly longer MACE-free survival compared with patients with transcoronary concentration gradients of miR-133a above the median in the overall population (A), in stable CAD subgroup (B), and in ACS subgroup (C). Statistical assessment was performed using the log-rank test.

circulation, neither miR-133a concentration in CVS nor in aorta showed any correlation with the cumulative incidence of MACE at follow-up (Table 3, A to C). In contrast, miR-133a transcoronary concentration gradients $>$ median were associated with a significantly higher incidence of MACE ($p < 0.001$; Table 3 A to C and Figure 1). Regarding individual events, we found that increased miR-133a transcoronary concentration gradients were positively associated with the incidence of death ($p = 0.001$) and target lesion revascularization for in-stent-restenosis ($p = 0.032$). Among the patients experiencing in-stent restenosis, one had received bare-metal stents and the others had received drug-eluting stents at the index procedure. In both subgroups, the positive associations between higher miR-133a transcoronary concentration gradients and incidence of MACE remained significant (stable CAD patients: $p = 0.011$; ACS patients: $p = 0.002$, Table 3 B to C and Figure 1) and was driven by a higher revascularization rate in stable CAD group (overall incidence of revascularization: $p = 0.037$, restenosis: $p = 0.026$, de novo revascularization: $p = 0.311$) and by a higher mortality in ACS group ($p = 0.017$).

Survival analysis showed a significantly better MACE-free survival in patients with transcoronary concentration gradients of miR-133a $<$ median compared with those with concentrations $>$ median both in the overall population ($p < 0.001$, Figure 2) and in patients with stable CAD and in ACS ($p = 0.026$ and $p = 0.007$, respectively, Figure 2). With regard to individual events, in stable CAD population, patients with miR-133a concentrations $>$ median showed a higher revascularization rate that was driven by a higher incidence of in-stent-restenosis ($p = 0.034$ and $p = 0.014$, respectively, Figure 3); in ACS patients, miR-133 transcoronary concentration gradients were related to mortality, whereas no significant effects were observed on need for revascularization ($p = 0.020$ and $p = 0.089$,

respectively, Figure 3). In the simple Cox regression analysis (Table 4), transcoronary concentration gradients of miR-133a were able to predict the incidence of MACE in the overall population and in patients with ACS ($p = 0.002$ and $p = 0.032$, respectively) but not in patients with stable CAD ($p = 0.181$). However, this significant association was lost upon adjustment for hs-TnT levels, suggesting that miR-133a transcoronary concentration gradients are unable to independently predict clinical events. To further investigate the association between circulating concentration of miR-133a and myocardial injury, we correlated plasma levels of this miR with hs-TnT levels, both in CVS and in the transcoronary circulation. No significant correlation between concentrations of miR-133a and hs-TnT was observed in patients with stable CAD (Figure 4), whereas a significant positive correlation was observed between miR-133a and hs-TnT levels in CVS and, more strikingly, between miR-133a and hs-TnT release into the coronary circulation in patients with ACS ($p = 0.001$ and $p = 0.002$, respectively, Figure 4).

To get better insights into the observed association between miR-133a and patients' outcome, we analyzed the baseline characteristics of our study population according to miR-133a transcoronary concentration gradients (Table 5). No significant difference was observed in the overall population. In patients with stable CAD, a significantly different New York Heart Association class was observed between patients with miR-133a transcoronary levels $<$ and $>$ median, but no difference was found when considering only the highest New York Heart Association classes (≥ 3). In patients presenting with ACS, transcoronary miR-133a levels were associated with higher hs-TnT levels at admission, confirming the correlation between miR-133a cardiac release and extent of cardiac damage, as well as with higher C-reactive protein levels.

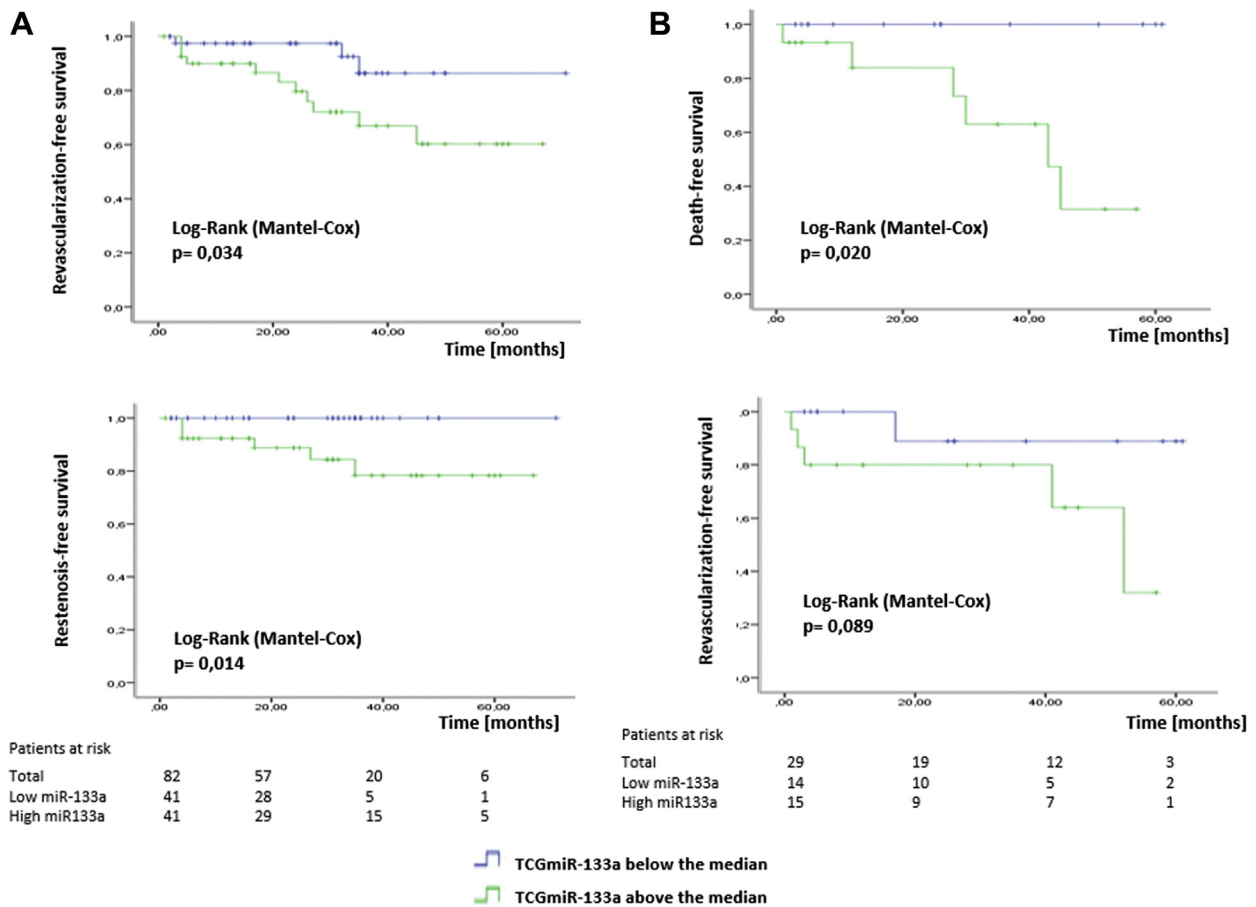


Figure 3. Impact of concentration gradients of miR-133a on individual adverse events. (A) In patients with stable CAD, transcoronary concentration gradients of miR-133a below the median were associated with a significantly longer event-free survival with regard to need of new revascularization and target lesion revascularization because of in-stent restenosis, compared with patients with transcoronary concentration gradients of miR-133a above the median; (B) in patients with ACS, transcoronary concentration gradients of miR-133a below the median were associated with a significantly longer death-free survival, whereas no association was observed with regard to the need of revascularization, compared with patients with transcoronary concentration gradients of miR-133a above the median. Statistical assessment was performed using the log-rank test.

Table 4

Predictors of MACE in simple and stepwise multiple Cox regression analysis

Variable	Overall Population		Stable CAD patients		ACS patients	
	Hazard ratio	P	Hazard ratio	p	Hazard ratio	p
Age	1,008 (0,971-1,045)	0,682	1,022 (0,973-1,074)	0,381	0,987 (0,932-1,037)	0,660
Diabetes	1,365 (0,618-3,018)	0,441	1, 204 (0,413-3,508)	0,734	1,716 (0,497-5,921)	0,393
LVEF	0,987 (0,953-1,022)	0,458	1,058 (0,984-1,137)	0,127	0,952 (0,897-1,011)	0,109
CRP	1,111 (0,968-1,276)	0,716	0,786 (0,335-1,884)	0,786	1,088 (0,944-1,255)	0,246
NT-pro-BNP	1,001 (1,00-1,001)	0,372	1 (0,999-1,002)	0,729	1 (0,999-1)	0,954
hs-TnT _[CVS]	1,001 (1,001-1,002)	<0,0001	1,005 (0,946-1,068)	0,878	1,001 (1,001-1,002)	0,013
hs-TnT _[TCG]	1,001 (1,001-1,002)	<0,0001	0,977 (0,840-1,136)	0,758	1,001 (1,001-1,002)	0,012
miR-133a _[TCG]	1,003 (1,001-1,004)	0,002	1,386 (0,859-2,238)	0,181	1,002 (1,001-1,004)	0,032
miR-133a _[TCG] adjusted for Δ hs-TnT _[TCG]	1,000 (0,997-1,003)	0,956	-	-	1,001 (1-1,001)	0,971
miR-133a _[TCG] adjusted for hs-TnT _[CVS]	1,001 (0,998-1,003)	0,565	-	-	1 (0,998-1,003)	0,799

ACS = acute coronary syndrome; CAD = coronary artery disease; CRP = C-reactive protein; CVS = coronary venous sinus; hs-TnT = high sensitivity troponin T; LVEF = left ventricular ejection fraction; NT-pro-BNP = N-terminal pro-B-type natriuretic peptide; TCG = transcoronary concentration gradient.

Predictors of MACE were identified by simple Cox regression analysis and tested in a stepwise multiple Cox regression analysis. Bold indicates statistical significance at $p < 0.05$.

Discussion

In this prospective study of 111 patients, we found that increased transcoronary concentration gradients of miR-133a

are associated with a higher incidence of MACE during long-term follow-up. In particular, the increased incidence of cumulative MACE was driven by a higher mortality rate in patients with ACS and by a higher occurrence of target lesion

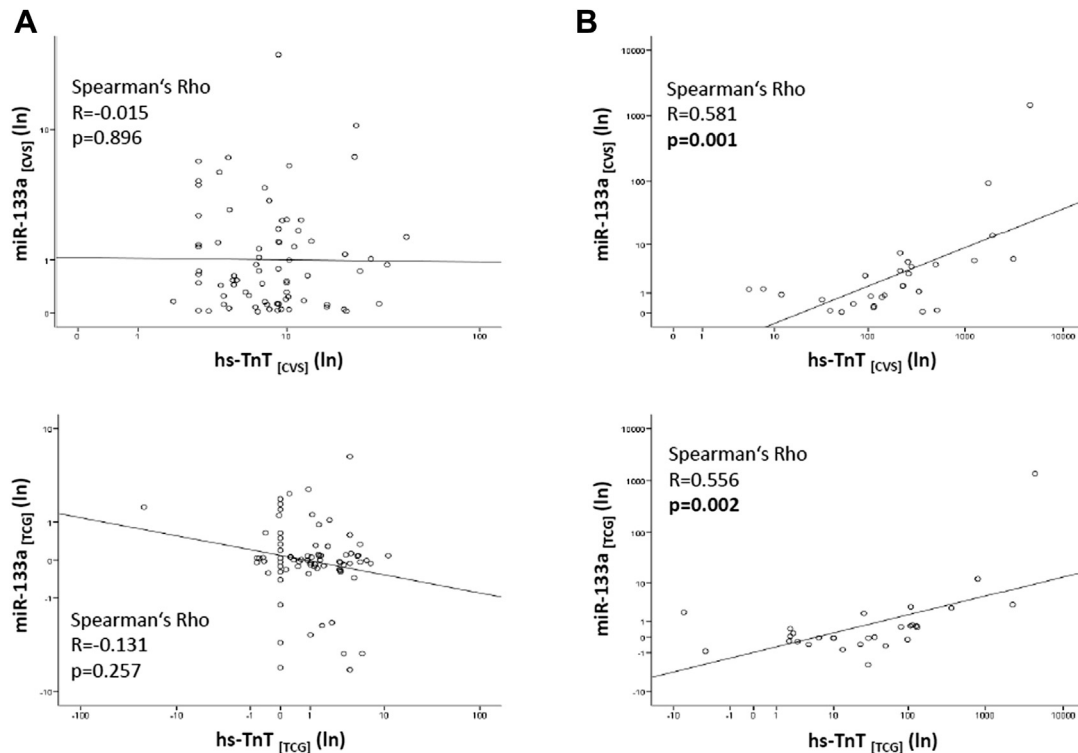


Figure 4. Correlations between plasma levels of miR-133a and hs-TnT. (A) In patients with stable CAD, no significant correlation was observed between plasma concentration of miR-133a and hs-TnT, nor in the coronary venous blood neither in the coronary circulation; (B) in patients with ACS, plasma concentrations of miR-133a and hs-TnT showed a significant positive correlation both in the coronary venous blood and as measured by transcoronary concentration gradients. The Spearman correlation was used to compare levels of miR-133a and hs-TnT after logarithmic transformation.

revascularization in patients with stable CAD. The increased transcoronary concentration gradients of miR-133a may be due to either a higher cardiac release or a reduced degradation of this miR during the transcoronary passage. Although it is impossible to discriminate between these 2 mechanisms in patients *in vivo*, several findings suggest that the increase of miR-133a levels more likely is caused by accentuated release rather than by reduced degradation. First, a reduced miR-degradation is hard to expect during myocardial injury, which increases the secretion of proteinases and ribonucleases^{18,19}; however, microvesicle-embedded miRs could be resistant to proteinase-associated degradation processes. Second, in case of generally reduced degradation, one would expect similar concentration patterns for different miRs, which is not the case; indeed, several experimental and clinical studies have demonstrated that miR-133a plasma concentrations are selectively increased both in patients with CAD²⁰ and in patients with ACS,⁴ and different miRs-concentration patterns were observed also in studies selectively investigating microvesicle-embedded miRs.²¹

In patients with ACS, miR-133a plasma concentrations are closely correlated with the extent of myocardial injury.^{6,10} In our ACS-population, we found that an increased cardiac release of miR-133a is closely correlated with the release of hs-TnT into the coronary circulation, indicating a significant relation between the release of this miR and extent of cardiac injury. This strict relation explains the observed significant association between higher transcoronary concentration gradients of miR-133a and cardiovascular outcome in our

patients with ACS. Indeed, when corrected for the release of hs-TnT, miR133a concentrations do no longer predict patients' outcome, indicating that increased miR133a release is secondary to myocardial damage.

In contrast, the mechanism underlying the increase of plasma levels of miR-133a observed in patients with stable CAD²⁰ is currently not fully understood. Of course, one cannot rule out that miR-133a elevation in these patients might be caused by a low grade of chronic myocardial injury. Nevertheless, this hypothesis is unlikely to affect the results in our population of stable patients, for several reasons. First, none of our stable patients showed an increase in hs-TnT levels above the well-established "gray area" of 50 pg/ml; therefore, none of them had hs-TnT levels diagnostic of *on-going* myocardial ischemia. Second, no difference in troponin levels was found between stable CAD patients with higher and lower miR-133a levels (Table 5). Third, well-established markers of myocardial wall stress and ischemia (N-terminal pro-B-type natriuretic peptide) and inflammation (C-reactive protein) or cardiac function and structure (left ventricular ejection fraction and left ventricular hypertrophy) did not significantly differ between stable CAD with higher and lower miR-133a levels (Table 5). Moreover, we did not observe any significant correlation between plasma concentrations of miR-133a and hs-TnT levels in the CVS or in the coronary circulation in patients with stable CAD and very-low hs-TnT levels. These results confirm our previous findings that very minute elevations of hs-TnT are not reflected by increases in circulating plasma levels of the

Table 5
Patients' characteristics with respect to miR133a transcoronary concentration gradients

Variable	Overall population			Stable CAD patients			ACS patients		
	miR-133 _[TCG]			miR-133 _[TCG]			miR-133 _[TCG]		
	< median (N=55)	> median (N=56)	P	< median (N=41)	> median (N=41)	P	< median (N=14)	> median (N=15)	P
Age	61,6±11,3	65,4±11,0	0,076	61,0±11,8	66,0±11,1	0,050	64,9±9,6	61,9±11,1	0,445
Sex (M/F)	42/13	47/9	0,476	32/9	32/9	1	10/4	15/0	0,042
Arterial Hypertension	45 (82%)	48 (86%)	0,616	32 (78%)	36 (88%)	0,379	13 (93%)	12 (80%)	0,598
Diabetes mellitus	12 (22%)	17 (30%)	0,388	8 (19%)	13 (32%)	0,312	4 (29%)	4 (27%)	1
Current Smoker	24 (44%)	19 (34%)	0,594	17 (41%)	15 (37%)	0,330	7 (50%)	4 (27%)	0,409
Previous AMI	21 (38%)	23 (41%)	0,847	17 (41%)	20 (49%)	0,657	3 (21%)	4 (27%)	1
Previous PCI	30 (54%)	32 (57%)	0,849	26 (63%)	29 (71%)	0,639	3 (21%)	4 (27%)	1
NYHA Class (0/I/II/III/IV)	10/13/21/9/2	11/26/11/7/1	0,089	8/9/14/8/2	8/22/5/6/0	0,017	3/3/7/1/0	2/5/6/1/1	0,783
-NYHA Class ≥ III	-11	-8	0,460	-10	-6	0,404	-1	-2	0,527
CCS Class (0/I/II/III/IV)	10/4/9/15/17	12/4/5/13/22	0,717	9/4/9/15/4	13/4/5/12/7	0,554	0/0/0/0/14	0/0/0/1/14	1
LV-EF (%)	52,7±11,3	52,8±10,6	0,913	52,5±11,2	56,4±7,4	0,419	50,9±12,0	42,6±11,5	0,093
Leukocytes	8,1±2,7	8,3±3,2	0,755	7,7±2,4	7,3±1,7	0,749	9,5±3,3	11,3±4,3	0,252
Creatinine	0,97±0,22	1,04±0,25	0,153	1,0±0,2	1,0±0,2	0,370	0,9±0,2	1,2±0,3	0,077
hs-Tn T _[CVS]	49,3±97,8	275,2±802,3	0,703	10,0±8,4	9,8±7,1	0,929	153,4±140,7	979,9±1323,3	0,020
NT-proBNP _[CVS]	483,4±677,4	537,8±771,4	0,649	312,4±377,1	346,8±394,5	0,620	938,7±1030,3	1056,4±1225,1	0,983
C-reactive-Protein	0,54±0,99	0,88±2,38	0,725	0,5±1,1	0,3±0,4	0,716	0,5±0,8	2,6±4,3	0,024
Cholesterol	181,6±40,5	189,0±52,1	0,43	178,1±36,9	192,6±54,3	0,183	191,8±49,5	178,5±45,6	0,483
-LDL-Cholesterol	-103,3±41,0	-110,1±40,7	0,38	-102,3±43,2	-107,6±40,9	0,508	-105,7±34,0	-120,1±41,0	0,468
Statin	38 (69%)	38 (68%)	1	31 (76%)	35 (85%)	0,404	6 (43%)	4 (27%)	0,695
ACE-I or ARB	43 (78%)	43 (77%)	1	33 (80%)	38 (93%)	0,194	9 (64%)	6 (40%)	0,449
Insulin	6 (11%)	4 (7%)	0,742	31 (76%)	35 (85%)	0,675	2 (14%)	2 (13%)	1
Antiplatelet therapy									
-ASS	49 (89%)	45 (80%)	0,441	35 (85%)	35 (85%)	0,441	12 (86%)	12 (80%)	1
-Clopidogrel/Ticagrelor/Prasugrel	38 (69%)	39 (70%)	0,870	26 (63%)	29 (71%)	0,820	10 (71%)	12 (80%)	0,682
Oral anticoagulation therapy	3 (5%)	2 (4%)	0,547	3 (7%)	2 (5%)	0,549	0 (0%)	0 (0%)	1

All p values represent comparisons between patients with miR-133a transcoronary concentration gradients below and above the median value. Comparisons between groups were performed with the Fisher's exact test or chi-square test for categorical variables and with Student's *t* test or Mann-Whitney *U* test for continuous variables. Bold indicates statistical significance at *p* < 0.05.

ACS = acute coronary syndrome; ACE-I = angiotensin-converting enzyme inhibitors; AMI = acute myocardial infarction; ARB = angiotensin receptor blocker; CAD = coronary artery disease; CCS = Canadian Cardiovascular Society; CVS = coronary venous sinus; hs-TnT = high sensitivity troponin T; LDL = low-density lipoprotein; LVEF = left ventricular ejection fraction; NT-pro-BNP = N-terminal pro-B-type natriuretic peptide; NYHA = New York Heart Association; PCI = percutaneous coronary intervention.

muscle-enriched miRs.¹⁰ Thus, the release of miR-133a did not reflect the presence of subclinical myocardial injury as measured by troponin release into the coronary circulation. A potential explanation for increase in miR-133a transcoronary concentration gradients in stable patients could be related to the recently established role of this miR in VSMCs function and phenotype.³ In fact, it is well known that VSMCs, unlike adult skeletal and cardiac muscle cells, retain a remarkable phenotypic plasticity and could undergo relevant changes in gene expression, contractility, extracellular matrix, and inflammatory proteins' production and cell cycle state, which represents one of the first steps in the development of atherosclerotic lesions.²² In this light, the significant association that we found between increased transcoronary concentration gradient of miR-133a and incidence of in-stent restenosis, although just at an observational level, appears to be remarkable because this latter is a pathological process in which the "phenotypic switch" of VSMCs from a synthetic to a proliferative state plays a pivotal role. Notably, we did not found any association between transcoronary concentration gradient of miR-133a and incidence of revascularization because of a de novo stenosis, suggesting a selective

involvement of this miR in the atherosclerosis progression in response to vascular injury, as also shown in the experimental models.³ If supported from mechanistic experimental studies as well as from larger clinical trials, our observational data could open intriguing perspectives in prevention of in-stent restenosis. First, it could allow the identification of patients at higher risk of in-stent restenosis, in which a stress-imaging test or a control coronary angiography, irrespective of symptoms, may be considered.¹³ Second, the modulation of miR-133a expression and function in coronary VSMCs could represent the basis for the development of miR-targeted drugs for the prevention of in-stent restenosis.

The present study has several limitations. First, it is a single-center study, and the sample size is small, which may make the interpretation of subgroup analyses for individual events challenging. Second, miR measurements currently are performed by PCR, which is affected by the lack of validated internal controls and makes miRs measurements sensitive for RNA extraction from blood samples and potential interindividual confounding factors, for example, heparin¹¹ and other medical therapies.²³ Nevertheless, the assessment of transcoronary concentration gradients of miR-133a may have

provided informative data even in a small cohort of patients, whereas the systemic concentration of circulating miR-133a might be influenced by several interindividual confounding factors, for example, antiplatelet therapy,²⁴ as well as the presence of atherosclerotic lesions in the noncoronary vasculature, explaining, at least in part, the lack of association between patients' outcome and circulating miR-133a levels in the coronary sinus (which most closely trace the systemic venous concentrations²⁵). Moreover, the need of CVS cannulation makes the measurements of miRs transcoronary concentration gradients very difficult in daily practice. As such, the present study should be primarily viewed as a proof-of-concept study addressing a potential pathophysiological role of circulating miRs in the coronary circulation. Finally, technical improvement in miRs measurement (e.g., use of hybridization technologies instead of PCR) may provide a path toward a more standardized application of assessing miRs levels as circulating biomarkers of cardiovascular diseases in clinical practice.

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Disclosures

The authors have no conflicts of interest to disclose.

- Chen X, Ba Y, Ma L, Cai X, Yin Y, Wang K, Guo J, Zhang Y, Chen J, Guo X, Li Q, Li X, Wang W, Zhang Y, Wang J, Jiang X, Xiang Y, Xu C, Zheng P, Zhang J, Li R, Zhang H, Shang X, Gong T, Ning G, Wang J, Zen K, Zhang J, Zhang CY. Characterization of microRNAs in serum: a novel class of biomarkers for diagnosis of cancer and other diseases. *Cell Res* 2008;18:997–1006.
- Latronico MV, Catalucci D, Condorelli G. Emerging role of microRNAs in cardiovascular biology. *Circ Res* 2007;101:1225–1236.
- Torella D, Iaconetti C, Catalucci D, Ellison GM, Leone A, Waring CD, Boicchio A, Vicinanza C, Aquila I, Curcio A, Condorelli G, Indolfi C. MicroRNA-133 controls vascular smooth muscle cell phenotypic switch in vitro and vascular remodeling in vivo. *Circ Res* 2011;109:880–893.
- Widera C, Gupta SK, Lorenzen JM, Bang C, Bauersachs J, Bethmann K, Kempf T, Wollert KC, Thum T. Diagnostic and prognostic impact of six circulating microRNAs in acute coronary syndrome. *J Mol Cell Cardiol* 2011;51:872–875.
- Zampetaki A, Willeit P, Tilling L, Drozdov I, Prokopi M, Renard JM, Mayr A, Weger S, Schett G, Shah A, Boulanger CM, Willeit J, Chwienczyk PJ, Kiechl S, Mayr M. Prospective study on circulating MicroRNAs and risk of myocardial infarction. *J Am Coll Cardiol* 2012;60:290–299.
- Eitel I, Adams V, Dieterich P, Fuernau G, de Waha S, Desch S, Schuler G, Thiele H. Relation of circulating MicroRNA-133a concentrations with myocardial damage and clinical prognosis in ST-elevation myocardial infarction. *Am Heart J* 2012;164:706–714.
- Pilbrow AP, Cordeddu L, Cameron VA, Frampton CM, Troughton RW, Doughty RN, Whalley GA, Ellis CJ, Yandle TG, Richards AM, Foo RS. Circulating miR-323-3p and miR-652: candidate markers for the presence and progression of acute coronary syndromes. *Int J Cardiol* 2014;176:375–385.
- Wang GK, Zhu JQ, Zhang JT, Li Q, Li Y, He J, Qin YW, Jing Q. Circulating microRNA: a novel potential biomarker for early diagnosis of acute myocardial infarction in humans. *Eur Heart J* 2010;31:659–666.
- Thygesen K, Alpert JS, Jaffe AS, Simoons ML, Chaitman BR, White HD, Writing Group on the Joint ESC/ACCF/AHA/WHF Task Force for the Universal Definition of Myocardial Infarction; Thygesen K, Alpert JS, White HD, Jaffe AS, Katus HA, Apple FS, Lindahl B, Morrow DA, Chaitman BA, Clemmensen PM, Johanson P, Hod H, Underwood R, Bax JJ, Bonow RO, Pinto F, Gibbons RJ, Fox KA, Atar D, Newby LK, Galvani M, Hamm CW, Uretsky BF, Steg PG, Wijns W, Bassand JP, Menasche P, Ravkilde J, Ohman EM, Antman EM, Wallentin LC, Armstrong PW, Simoons ML, Januzzi JL, Nieminen MS, Gheorghiade M, Filippatos G, Luepker RV, Fortmann SP, Rosamond WD, Levy D, Wood D, Smith SC, Hu D, Lopez-Sendon JL, Robertson RM, Weaver D, Tendera M, Bove AA, Parkhomenko AN, Vasilieva EJ, Mendis S. Guidelines ESC/EF: third universal definition of myocardial infarction. *Eur Heart J* 2012;33:2551–2567.
- De Rosa S, Fichtlscherer S, Lehmann R, Assmus B, Dimmeler S, Zeiher AM. Transcoronary concentration gradients of circulating microRNAs. *Circulation* 2011;124:1936–1944.
- Boeckel JN, Thome CE, Leistner D, Zeiher AM, Fichtlscherer S, Dimmeler S. Heparin selectively affects the quantification of microRNAs in human blood samples. *Clin Chem* 2013;59:1125–1127.
- Authors/Task Force members; Windecker S, Kolh P, Alfonso F, Collet JP, Cremer J, Falk V, Filippatos G, Hamm C, Head SJ, Juni P, Kappetein AP, Kastrati A, Knuuti J, Landmesser U, Laufer G, Neumann FJ, Richter DJ, Schauerte P, Sousa Uva M, Stefanini GG, Taggart DP, Torracca L, Valgimigli M, Wijns W, Witkowski A. 2014 ESC/EACTS guidelines on myocardial revascularization: the Task Force on myocardial revascularization of the European Society of Cardiology (ESC) and the European Association for Cardio-Thoracic Surgery (EACTS) Developed with the special contribution of the European Association of Percutaneous Cardiovascular Interventions (EAPCI). *Eur Heart J* 2014;35:2541–2619.
- Task Force M, Montalescot G, Sechtem U, Achenbach S, Andreotti F, Arden C, Budaj A, Bugiardini R, Crea F, Cuisset T, Di Mario C, Ferreira JR, Gersh BJ, Gitt AK, Hulot JS, Marx N, Opie LH, Pfisterer M, Prescott E, Ruschitzka F, Sabate M, Senior R, Taggart DP, van der Wall EE, Vrints CJ, Guidelines ESC Committee for Practice Guidelines; Zamorano JL, Achenbach S, Baumgartner H, Bax JJ, Bueno H, Dean V, Deaton C, Erol C, Fagard R, Ferrari R, Hasdai D, Hoes AW, Kirchhof P, Knuuti J, Kolh P, Lancellotti P, Linhart A, Nihoyannopoulos P, Piepoli MF, Ponikowski P, Simes PA, Tamargo JL, Tendera M, Torbicki A, Wijns W, Windecker S, Document R, Knuuti J, Valgimigli M, Bueno H, Claeys MJ, Donner-Banzhoff N, Erol C, Frank H, Funck-Brentano C, Gaemperli O, Gonzalez-Juanatey JR, Haimos M, Hasdai D, Husted S, James SK, Kervinen K, Kolh P, Kristensen SD, Lancellotti P, Maggioni AP, Piepoli MF, Pries AR, Romeo F, Ryden L, Simoons ML, Simes PA, Steg PG, Timmis A, Wijns W, Windecker S, Yildirim A, Zamorano JL. 2013 ESC guidelines on the management of stable coronary artery disease: the Task Force on the management of stable coronary artery disease of the European Society of Cardiology. *Eur Heart J* 2013;34:2949–3003.
- Task Force on the management of ST-segment elevation acute myocardial infarction of the European Society of Cardiology (ESC); Steg PG, James SK, Atar D, Badano LP, Blomstrom-Lundqvist C, Borger MA, Di Mario C, Dickstein K, Ducrocq G, Fernandez-Aviles F, Gershlick AH, Giannuzzi P, Halvorsen S, Huber K, Juni P, Kastrati A, Knuuti J, Lenzen MJ, Mahaffey KW, Valgimigli M, van 't Hof A, Widimsky P, Zahger D. ESC Guidelines for the management of acute myocardial infarction in patients presenting with ST-segment elevation. *Eur Heart J* 2012;33:2569–2619.
- Roffi M, Patrono C, Collet JP, Mueller C, Valgimigli M, Andreotti F, Bax JJ, Borger MA, Brotons C, Chew DP, Gencer B, Hasenfuss G, Kjeldsen S, Lancellotti P, Landmesser U, Mehilli J, Mukherjee D, Storey RF, Windecker S, Baumgartner H, Gaemperli O, Achenbach S, Agewall S, Badimon L, Baigent C, Bueno H, Bugiardini R, Carerj S, Casselman F, Cuisset T, Erol C, Fitzsimons D, Halle M, Hamm C, Hildick-Smith D, Huber K, Iliodromitis E, James S, Lewis BS, Lip GY, Piepoli MF, Richter D, Rosemann T, Sechtem U, Steg PG, Vrints C, Luis Zamorano J. 2015 ESC guidelines for the management of acute coronary syndromes in patients presenting without persistent ST-segment elevation: Task Force for the management of acute coronary syndromes in patients presenting without Persistent ST-Segment Elevation of the European Society of Cardiology (ESC). *Eur Heart J* 2016;37:267–315.
- Wykrzykowska JJ, Garg S, Onuma Y, de Vries T, Goedhart D, Morel MA, van Es GA, Buszman P, Linke A, Ischinger T, Klaus V, Corti R, Eberli F, Wijns W, Morice MC, di Mario C, van Geuns RJ, Juni P, Windecker S, Serruys PW. Value of age, creatinine, and ejection fraction (ACEF score) in assessing risk in patients undergoing percutaneous coronary interventions in the 'All-Comers' LEADERS trial. *Circ Cardiovasc Interv* 2011;4:47–56.

17. Eagle KA, Lim MJ, Dabbous OH, Pieper KS, Goldberg RJ, Van de Werf F, Goodman SG, Granger CB, Steg PG, Gore JM, Budaj A, Avezum A, Flather MD, Fox KA; GRACE Investigators. A validated prediction model for all forms of acute coronary syndrome: estimating the risk of 6-month postdischarge death in an international registry. *JAMA* 2004;291:2727–2733.
18. Webb CS, Bonnema DD, Ahmed SH, Leonardi AH, McClure CD, Clark LL, Stroud RE, Corn WC, Finklea L, Zile MR, Spinale FG. Specific temporal profile of matrix metalloproteinase release occurs in patients after myocardial infarction: relation to left ventricular remodeling. *Circulation* 2006;114:1020–1027.
19. Sznajd J, Magdon M, Naskalski JW, Uracz R, Wojcikiewicz O. Serum ribonuclease activity in acute myocardial infarction. *Cor Vasa* 1981;23: 241–247.
20. Fichtlscherer S, De Rosa S, Fox H, Schwietz T, Fischer A, Liebetrau C, Weber M, Hamm CW, Roxel T, Muller-Ardogan M, Bonauer A, Zeiher AM, Dimmeler S. Circulating microRNAs in patients with coronary artery disease. *Circ Res* 2010;107:677–684.
21. Jansen F, Yang X, Proebsting S, Hoelscher M, Przybilla D, Baumann K, Schmitz T, Dolf A, Endl E, Franklin BS, Sinning JM, Vasa-Nicotera M, Nickenig G, Werner N. MicroRNA expression in circulating microvesicles predicts cardiovascular events in patients with coronary artery disease. *J Am Heart Assoc* 2014;3:e001249.
22. Owens GK, Kumar MS, Wamhoff BR. Molecular regulation of vascular smooth muscle cell differentiation in development and disease. *Physiol Rev* 2004;84:767–801.
23. Hoefer IE, Steffens S, Ala-Korpela M, Back M, Badimon L, Bochaton-Piallat ML, Boulanger CM, Caligiuri G, Dimmeler S, Egidio J, Evans PC, Guzik T, Kwak BR, Landmesser U, Mayr M, Monaco C, Pasterkamp G, Tunon J, Weber C; ESC Working Group Atherosclerosis and Vascular Biology. Novel methodologies for biomarker discovery in atherosclerosis. *Eur Heart J* 2015;36: 2635–2642.
24. Willeit P, Zampetaki A, Dudek K, Kaudewitz D, King A, Kirkby NS, Crosby-Nwaobi R, Prokopi M, Drozdov I, Langley SR, Sivaprasad S, Markus HS, Mitchell JA, Warner TD, Kiechl S, Mayr M. Circulating microRNAs as novel biomarkers for platelet activation. *Circ Res* 2013;112:595–600.
25. Boeckel JN, Reis SM, Leistner D, Thome CE, Zeiher AM, Fichtlscherer S, Keller T. From heart to toe: heart's contribution on peripheral microRNA levels. *Int J Cardiol* 2014;172: 616–617.